SAVEETHA SCHOOL OF ENGINEERING

OPERATING SYSTEMS

ASSIGNMENT | CO1

EVALUATE AND JUSTIFY THE OPERATING-SYSTEM DESIGN OF A SMART HOSPITAL

Student Details	Information
Name	S. Maruthu Pandi
Register Number	192511162
Department	B.E. Computer Science and Engineering (Cyber Security)
Course	CSA0421 – Operating Systems
Assignment	Assignment 1 (Tasks 4)

TASK 4 – EVALUATE THE SERVICES AN OPERATING SYSTEM MUST PROVIDE

4.1 OS service requirements by environment

OS service	Monitors	Tablets / desktops	Database / imaging
User interface	Minimal status display and local alarm controls only.	Full touch or graphical interface; accessibility and usability.	Limited administrator console; service interfaces and monitoring.
Program execution	Only verified sensor, alarm and communication tasks.	Clinical apps, managed background apps and productivity tools.	Database engine, imaging software, backup and monitoring processes.
I/O operations	Deterministic sensor sampling, alarm output and bounded communication.	Touchscreen, display, wireless, storage and input devices.	High-throughput block I/O, network I/O and imaging devices.
File system	Small configuration/log storage with safe writes; avoid unnecessary general-purpose filesystems.	Encrypted local files and managed application storage.	Journaled filesystem, controlled mounts, database files, logs and backup storage.
Communication	Small, allow-listed device protocol; non-critical telemetry.	Secure Wi-Fi / LAN, application APIs and device management.	Database replication, client access, cluster heartbeat and partner APIs.
Error detection	Sensor range checks, driver error handling, watchdog and alarm self-test.	Application crash reporting, OS health and endpoint management.	Hardware health, disk checks, database integrity, replication and fail-over monitoring.

OS service	Monitors	Tablets / desktops	Database / imaging
Resource allocation	Fixed priorities, bounded memory and CPU budgets.	Fair scheduling, power-aware CPU and memory allocation.	CPU, memory, disk and network scheduling with workload controls.
Accounting	Minimal event counters and fault logs.	Device compliance, access and usage logs.	Audit trails, performance accounting, capacity and transaction records.
Protection / security	Secure boot where supported, signed updates, isolated tasks and restricted device access.	Authentication, encryption, screen lock, app controls and remote wipe.	RBAC, least privilege, network segmentation, encryption, audit and backup protection.

4.2 Services that can be removed or reduced on a bedside monitor

To make the monitor smaller and more deterministic, remove services that are not needed for its clinical function. Examples include a general-purpose browser, office applications, consumer multimedia services, print spooler, broad software package manager, unnecessary discovery protocols, remote desktop server, general-purpose multi-user shell and large desktop GUI. A full general-purpose filesystem can be replaced with a small, carefully designed configuration and log store if the hardware/software design permits it.

Do not remove the services required for safety: deterministic task scheduling, timer and interrupt handling, sensor and alarm drivers, bounded memory management, watchdog support, self-test, fault logging, secure update verification and authenticated communication. “Small” must not mean “unprotected.”

4.3 Essential services on the database server

- Reliable process execution and scheduling for the DBMS, replication, backup and health-monitoring agents.
- Filesystem and block-device management with correct ordering, durability and recovery behaviour.

- Network communication for clients, replication, cluster heartbeat and secure partner interfaces.
- Memory management and resource allocation so database cache and transaction workloads receive predictable resources.
- Protection services: user/group identity, file permissions, mandatory access controls where appropriate, encrypted storage and secure network configuration.
- Error detection, audit/accounting, time synchronization, backup, restore and fail-over support.

4.4 Protecting one patient's records from another user's programs

Protection must exist at more than one layer. At the OS layer, separate service accounts, file permissions, process address spaces and mandatory access controls prevent an ordinary process from reading database files or another process's memory. At the database layer, authenticated users receive role-based permissions and row/record-level restrictions where required. At the application layer, the logged-in clinical identity and care relationship determine which records can be viewed. Encryption protects data in transit and at rest, while audit logs record who accessed or changed a record and when.

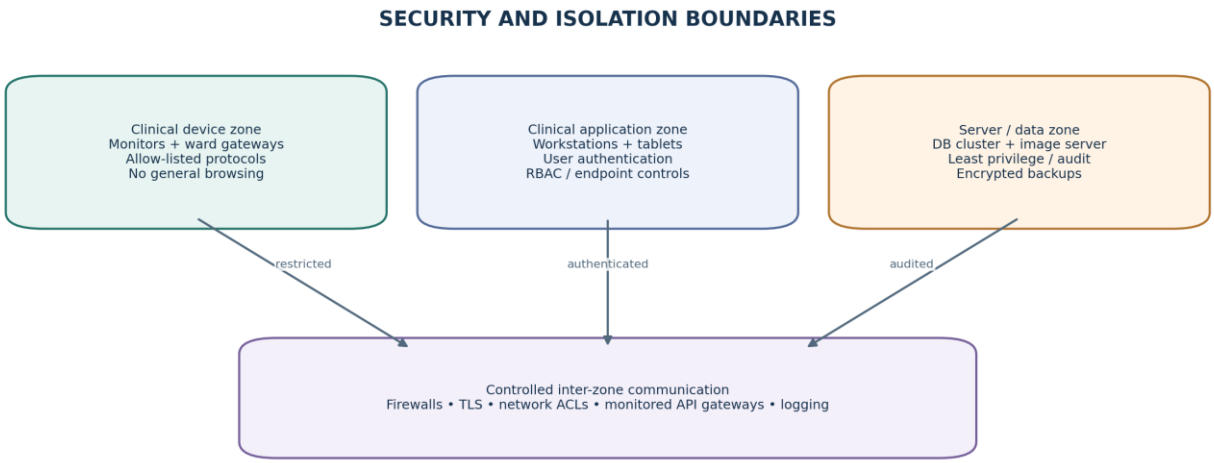


Figure 3. Network and security zones limit lateral movement and constrain which systems can communicate with the record service.

4.5 Alternative: full desktop OS on tablets

A full desktop OS would add background services, broader driver support, additional update components and more general-purpose applications. That increases storage requirements and

often increases RAM use. More services can increase boot time and background CPU/network activity. Higher idle power use can shorten battery life, which is important for a tablet used during a complete shift. Exact overhead cannot be stated without testing the chosen OS image and hardware, but the direction of the trade-off is clear.

A reduced handheld OS is preferred because its lifecycle and power-management model are designed for mobile devices. Hospital management tools can still enforce encryption, application control, authentication, patching and remote wipe without installing every desktop service.